

Maimoona Khilji

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Research Profile

Data Scientist and researcher with 3+ years of experience in machine learning, deep learning, and technical education, with research interests in multimodal AI, trustworthy and explainable AI, and large language models. Currently completing an M.S. in Data Science, with research on multimodal seizure detection from wearable biosignals using early fusion, deep learning, and explainable AI. Previous research includes NLP, healthcare AI, wearable AI, medical image classification, and retrieval-augmented generation.

Research Interests

Large Language Models	Multimodal AI	Trustworthy AI	Explainable AI	NLP
Deep Learning	Healthcare AI	Computer Vision	Wearable AI	RAG

Education

2025 - Present	Master of Science in Data Science Institute of Management Sciences, Pakistan	CGPA: 3.94 / 4.0
2019 - 2023	Bachelor of Science in Data Science Institute of Management Sciences, Pakistan	CGPA: 3.91 / 4.0 Distinction Holder

Relevant Coursework: Advanced Machine Learning, Artificial Intelligence, Natural Language Processing, Pattern Recognition, Data Mining, Statistics, Linear Algebra, Stochastic Processes, Big Data Analytics, and Database Systems.

Research Experience

M.S. Research	Multimodal Seizure Detection from Wearable Biosignals Institute of Management Sciences, Pakistan	2026 – Present
- Developing automated seizure detection using multimodal wearable biosignals from the SeizeIT2 dataset. - Investigating early-fusion approaches for integrating multimodal physiological signals. - Applying deep learning for automated seizure detection. - Incorporating explainable AI to improve model interpretability and transparency.		

Selected Research Projects

- Monitoring and Identifying Stress Using Wearable Health-Band Device (FYP)**
 - Developed an end-to-end stress detection system using physiological signals collected through a wearable health-band device.
 - Implemented an LSTM-based deep learning pipeline in Python, achieving 95.5% accuracy, with a Streamlit interface.

- **Explainable ML for Early Depression Detection in Students**
 - Compared Random Forest, SVM, KNN, XGBoost, AdaBoost, Logistic Regression, and MLP for early depression detection.
 - Applied SHAP, LIME, and DiCE to investigate model predictions and improve interpretability.
 - Logistic Regression achieved the strongest performance among the evaluated models.
- **Classification of Oral Squamous Cell Carcinoma**
 - Analyzed histopathological images for oral squamous cell carcinoma classification.
 - Compared VGG16, VGG19, and InceptionV3 architectures.
 - VGG19 demonstrated the strongest performance among the evaluated models.
- **Stress Detection Using Face Images**
 - Conducted a comparative study of SVM, Decision Tree, Random Forest, CNN, and VGG16 for stress detection from facial images.
 - Compared traditional machine learning and deep learning approaches.
 - VGG16 achieved the highest reported accuracy of 92%.
- **Transformer-Based NLP for Urdu Hate Speech Detection**
 - Evaluated Urdu-BERT, RoBERTa, XLM-R, and mBERT for Urdu hate speech detection in a low-resource setting.
 - Compared models using standard classification metrics; RoBERTa-Urdu-Small achieved 84% accuracy.
- **RAG-Based Document Question Answering System**
 - Developed a retrieval-augmented generation system designed to answer questions grounded exclusively in a provided document using Python, LangChain, Gemini 2.5 Flash, and Gradio.
 - Developed a separate AI-assisted prototype using Google AI Studio and explored trade-offs between rapid AI-assisted prototyping, transparency, customization, and control.

Professional Experience

Feb 2024 - Present

Lab Instructor | Institute of Management Sciences

- Conduct labs in Data Visualization, Data Warehousing & BI, Big Data Analytics, Data Mining, and AI using Python, SQL, PySpark, and Power BI.
- Support project-based learning using real-world datasets and applied analytics workflows.

Jan 2024 - Apr 2026

Data Science Trainer | Atomcamp

- Delivered practical Data Science training in Python, SQL, Excel, and Power BI to 2,000+ students.
- Designed project-based exercises covering data analysis, visualization, databases, BI, and machine learning using real-world datasets.

Jan 2026 - May 2026

BI Instructor | KPITB Training Program

- Delivered hands-on BI training and mentored participants in developing end-to-end solutions using real-world datasets.
- Covered data preparation, dashboard development, visualization, and data-driven decision-making.

Technical Skills

Machine Learning & Deep Learning	Scikit-learn, TensorFlow, Keras, PyTorch, NumPy, Pandas
Explainable & Trustworthy AI	SHAP, LIME, DiCE, Grad-CAM
Hardware & IoT	Arduino, Wemos D1 Mini, Sensors, IoT, Proteus, CupCarbon
Programming	Python, R, SQL, Java
Natural Language Processing	NLTK, Transformers, LangChain, RAG
Computer Vision	OpenCV, CNNs, VGG16, VGG19, InceptionV3
Databases	MySQL, Oracle, SQL Server, Google BigQuery
Big Data & Data Engineering	PySpark, Hive, MapReduce, Azure Databricks, Azure Data Lake
Data Visualization & BI	Power BI, Tableau, Matplotlib, Seaborn, Plotly, Dash, Excel
Application & Web Development	Streamlit, Gradio, Flask, HTML5, CSS, JavaScript, Vercel

Awards & Achievements

- Gold Medal for Academic Excellence, Institute of Management Sciences.
- Huawei Seeds For The Future PROGRAMME 2022 - Selected as the only candidate from Peshawar.
- Straight-A+ Academic Record throughout formal education.
- Speaker at tech events focused on Youth and Women empowerment.

Scholarship

2025-Present	Merit-Based Scholarship at Master’s level
2019-23	Merit-Based Partial Scholarship at Bachelor’s level
2017-19	Merit-Based Fully Funded Scholarship at College level
2007-17	Merit-Based Partial Scholarship at School level

Certificates

Microsoft	Microsoft Certified: Azure Data Fundamentals	2022
Microsoft	Microsoft Certified: Azure Fundamentals	2022
Databricks	AI Agent Fundamentals	2025
IBM	Machine Learning with Python	2025
IBM	Deep Learning & Neural Networks with Keras	2025
Google	Google AI Essentials	2024

Soft Skills

Collaboration	Active Learner	Problem Solving	Critical Thinking
Creativity	Time management	Adaptability	Detail-Oriented

Languages

English	Urdu	Pashto	Hindko
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